

Rayan Ahsan

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EDUCATION

University of Toronto

Toronto, ON

B.A.Sc. in Computer Engineering, Minor in Artificial Intelligence and Business Engineering Sep. 2023 – Apr. 2028

- **Relevant Coursework:** Data Structures and Algorithms, Object Oriented Programming, Probability and Statistics, Databases, Project Management (Agile/SDLC).

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, C++, SQL

Frameworks: React, Next.js, Node.js, Express, Flask, Django, PostgreSQL, MongoDB

Developer Tools: Git, Docker, Linux/Unix, AWS, Google Cloud Platform, Azure, Vercel

Libraries: Redux Toolkit, Material UI, Tailwind CSS, Recharts, Axios, Firebase, Prisma ORM

EXPERIENCE

ML Developer, Project: Aircraft Anomaly Detection

Sep. 2024 – April 2025

UofT Machine Intelligence Student Team

Toronto, ON

- Designed **STFT-based signal feature-extraction pipeline** and **spectral anomaly detection** for **50K+ aircraft sensor signals**, improving anomaly detection accuracy by **27%**.
- Trained and fine-tuned **ConvVAE** on real + synthetic data, reducing false positives by **35%**.
- Deployed anomaly scoring for predictive maintenance, extending component life by **20+ cycles**.

PROJECTS

RayStock – Inventory Management Web App | *Next.js, Redux Toolkit, PostgreSQL, Node.js, Prisma*

- Developed a full-stack inventory management system with a responsive **Next.js + Tailwind CSS** dashboard.
- Implemented **Redux Toolkit Query** for optimized state management and efficient data fetching.
- Built backend APIs with **Node.js + Prisma ORM** and connected to a **PostgreSQL** relational database.
- Integrated **Recharts** and **MUI Data Grid** for advanced data visualization and reporting.

SpeechBot – Real-Time Transcription & Translation Assistant | *React, Tailwind CSS, Node.js, Express*

- Implemented **real-time + offline transcription** using **OpenAI Whisper-tiny.en**, enabling accurate speech-to-text conversion.
- Integrated **NLLB-200 multilingual translation**, supporting seamless translation across multiple languages.
- Designed a responsive **React + Tailwind CSS** interface with support for both live recording and MP3 uploads.
- Added export functionality to save results as **PDF/TXT files**, enhancing accessibility and usability.
- Built backend with **Node.js + Express**, handling file uploads, ML inference, and integration with front-end.

Pokedex App | *React, TypeScript, Firebase, Firestore, Axios*

- Built an interactive **Pokedex web application** allowing users to search, filter, and manage Pokémon lists.
- Implemented secure authentication with **Firebase Auth (Google sign-in)** and persistent storage via **Firestore**.
- Fetched Pokémon data using **Axios API calls**, enabling dynamic UI rendering with React Router navigation.
- Delivered a fully responsive UI with custom SCSS styling and reusable components.

TripSync – Tourist Mapping Platform | *C++, OpenStreetMap API, A* Pathfinding*

- Designed a dynamic mapping platform with subway line rendering, **POI filtering**, and real-time route visualization.
- Optimized user experience with **0.2s zoom/pan**, **0.1s subway rendering**, and achieved **92% user satisfaction**.
- Implemented **multi-head A* search algorithm** for efficient route planning across multiple transit lines.

CERTIFICATES

Microsoft Certified: Azure AI Fundamentals

Aug. 2024

MATLAB Certified

Nov. 2023

IBM Applied AI Professional Certificate

Jan. 2022